CSSL Tornado Treefall Pattern Analysis Software User Manual

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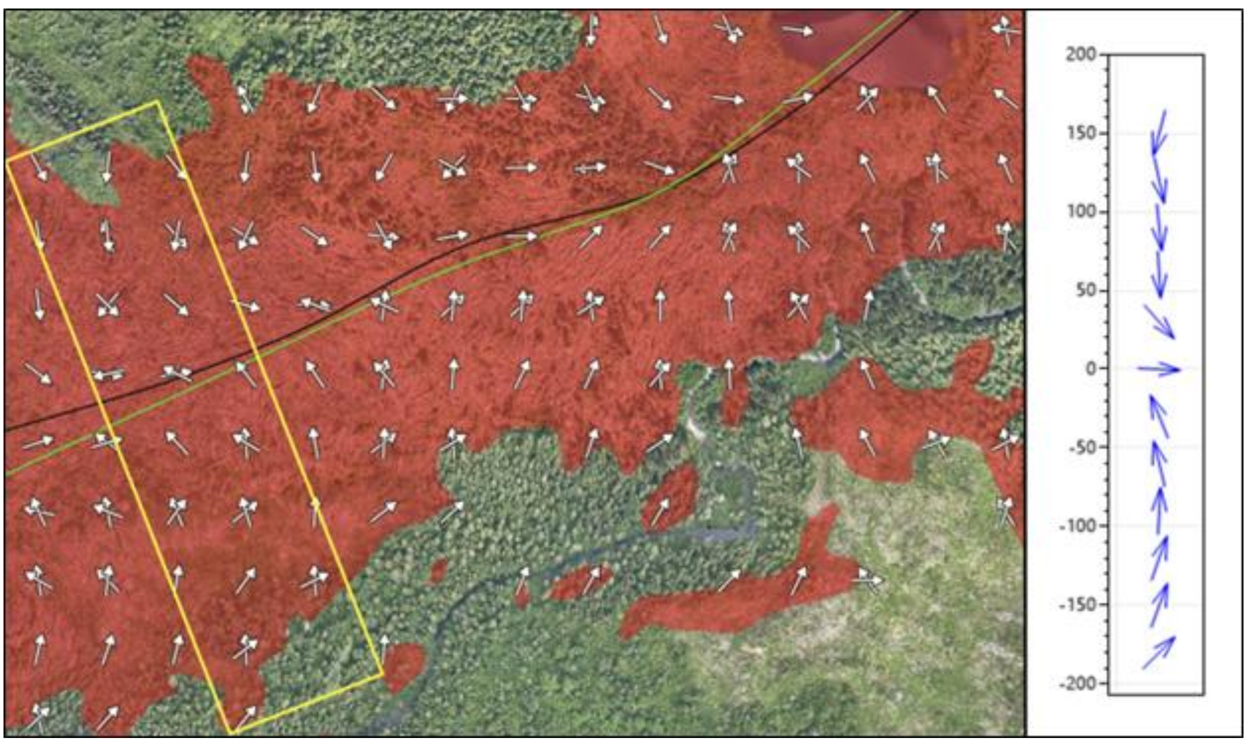

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Introduction

In Canada and many other parts of the world, often tornadoes travel through rural areas where trees are one of, if not the only significant damage indicator. This software is designed for the forensic analysis of tornado treefall patterns left behind by a tornado travelling through a forested area in order to estimate the tornado's maximum windspeed among other parameters. Moreover, this software is designed to be used in conjunction with analysis estimating the tornado's average translational speed, and the average critical tree failure speed of the damaged trees. Although treefall pattern matching has been used in urban areas, this software was designed for use in forested areas with clear treefall patterns and thousands of downed trees.

Estimates for the following are needed as prerequisites before this software can be used:

- Treefall directions (either individual or area-averaged with work)
- Tornado convergence line (path of treefall convergence)
- Average tornado translational speed (V_s) and uncertainty range
- Average critical tree failure speed (V_c) and uncertainty range

It is highly recommended that you read our paper for more information on the specific implementation details and analysis and consider also reading some of the listed previous literature shown in the paper's Table 2. This user manual will generally assume that you have read our paper.

Draft Paper

In order to detect trees and determine their treefall directions, CSSL utilizes an automated tree tagging software which can be downloaded from our GitHub:

https://github.com/Canadian-Severe-Storms-Laboratory/Tree_Tagger_ArcGIS_Add_On

Paper

Note: As of August 2025, this software is still under development. Features are bound to be added or changed, and there are almost certainly bugs in the software. If you're running into problems, see the "Troubleshooting" sections for help. If you're still stuck, feel free to email me at dbutt7@uwo.ca.

Recommended System Requirements

This software is designed as an add-in for ESRI's ArcGIS Pro desktop application. Moreover, this software is computationally intensive and optimized for multicore CPUs. As such, this software was designed for use on modern workstations with high CPU core counts. Below is a list of recommended minimum PC specs.

Note: If your PC can comfortably run ArcGIS Pro while loading large amounts of image and data files, then you should be fine to run this software, it might just run slower.

Recommended Specs

- Windows 10/11 (currently required for ArcGIS Pro)
- 16 GB RAM or more
- Intel i7 or AMD r7 from 2020 or later (e.g. i7 11700k / r7 5800x or better)
- Fast SSD storage space

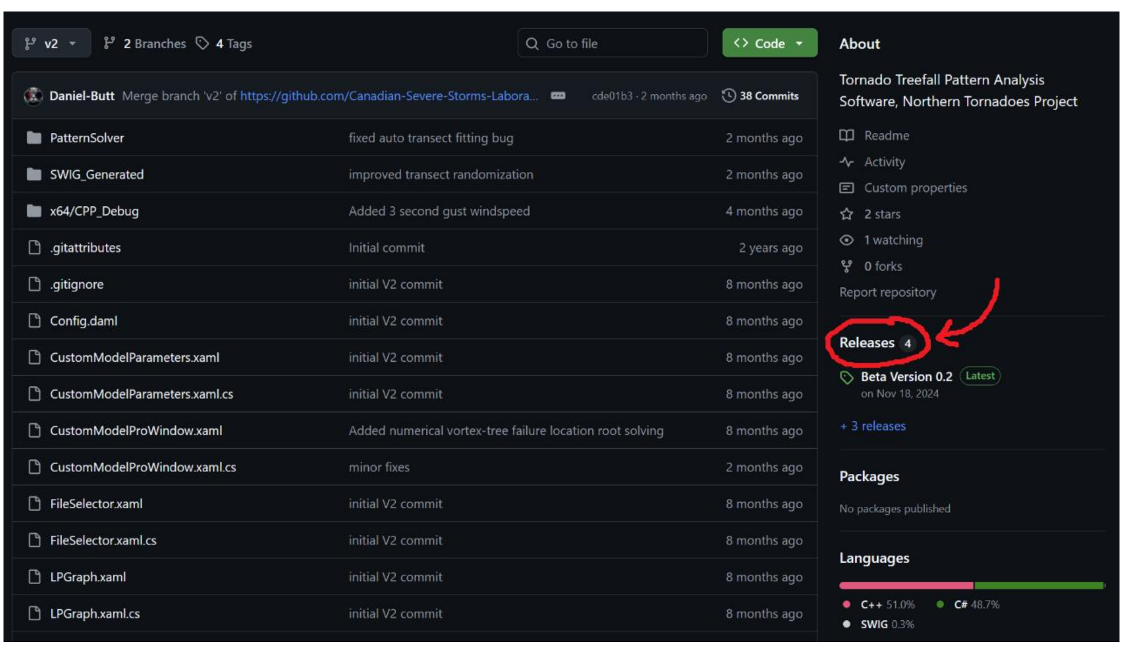
Installation

1. Install (or update to) the latest version of ArcGIS Pro (3.3+), then close ArcGIS Pro.

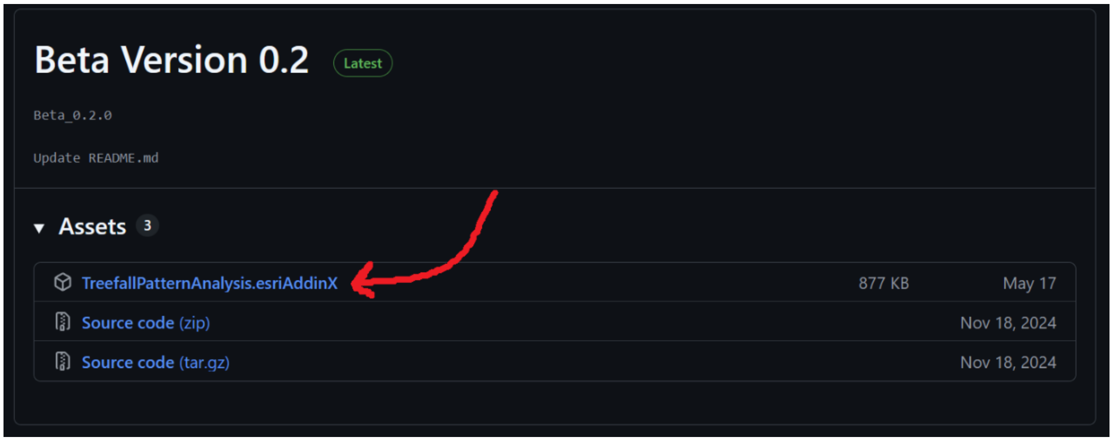
2. The software and this user manual can be downloaded from our GitHub page:

https://github.com/Canadian-Severe-Storms-Laboratory/Treefall_Pattern_Analysis

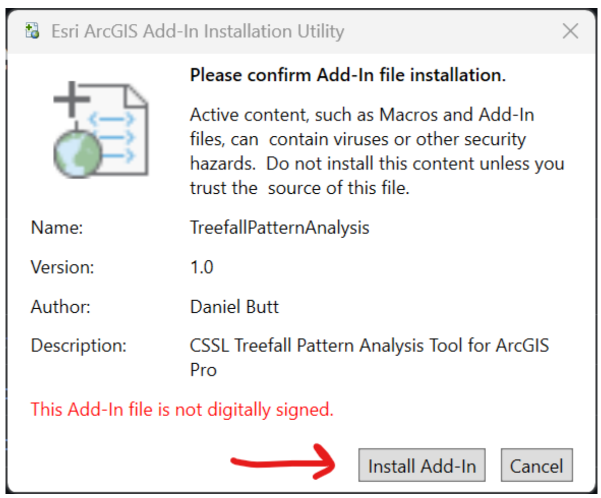
3. Once you're on the GitHub page, click on the "Releases" tab on the right side.



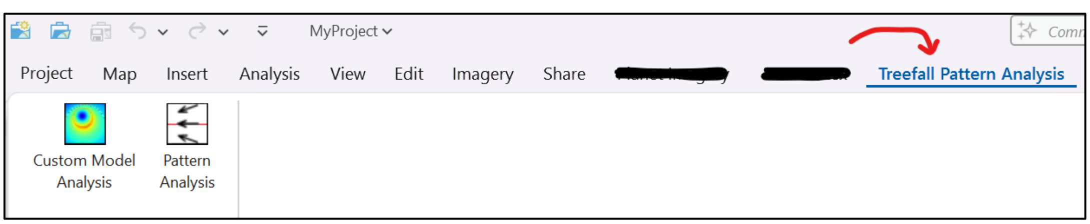
4. Click to download the latest version of the software by selecting the "TreefallPatternAnalysis.esriAddinX" file



5. Once downloaded, click to run the ".esriAddinX" file and click "Install Add-in".



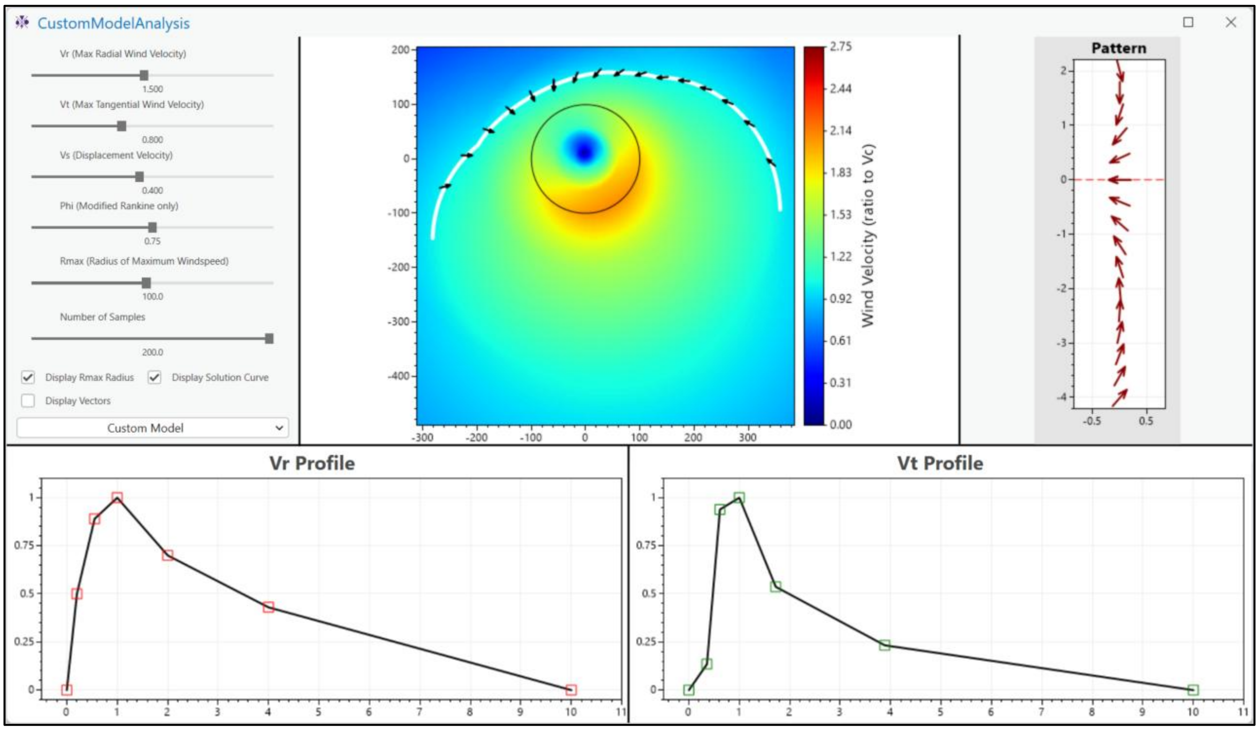
6. Open ArcGIS Pro (if open already, close and reopen ArcGIS Pro). Once you've opened a project you should now see the following tab in the header ribbon.



Clicking the available buttons ("Custom Model Analysis" and "Pattern Analysis") will open the respective software module. "Pattern Analysis" is the main software used for analyzing treefall patterns. "Custom Model Analysis" is used for visualizing and comparing different vortex models and the corresponding treefall patterns they can create.

Custom Model Analysis

This module of the software is primarily used for analyzing different vortex models and how changing different parameters can affect the corresponding treefall pattern. It is recommended to play around with this module to get a sense of what treefall patterns are plausible.



The panel on the left side contains various sliders and toggles which can be adjusted to change the parameters of the model. The graph in the middle displays a heatmap/wind field of the vortex model, along with the solution curve (vortex-tree failure location equation) and vectors which make up the treefall pattern. The corresponding treefall pattern is displayed in the graph on the right. The bottom two graphs show the radial (V_r) and tangential (V_t) profiles of the custom vortex model. All graphs can be adjusted by dragging or scrolling with the mouse. The custom profile graphs can be adjusted by dragging the nodes. Pushing the “d” or “n” key while dragging a node will delete it or add a new node next to it respectively. The other standard vortex models can also be selected from the drop-down menu under “Custom Model”.

Other Useful Notes:

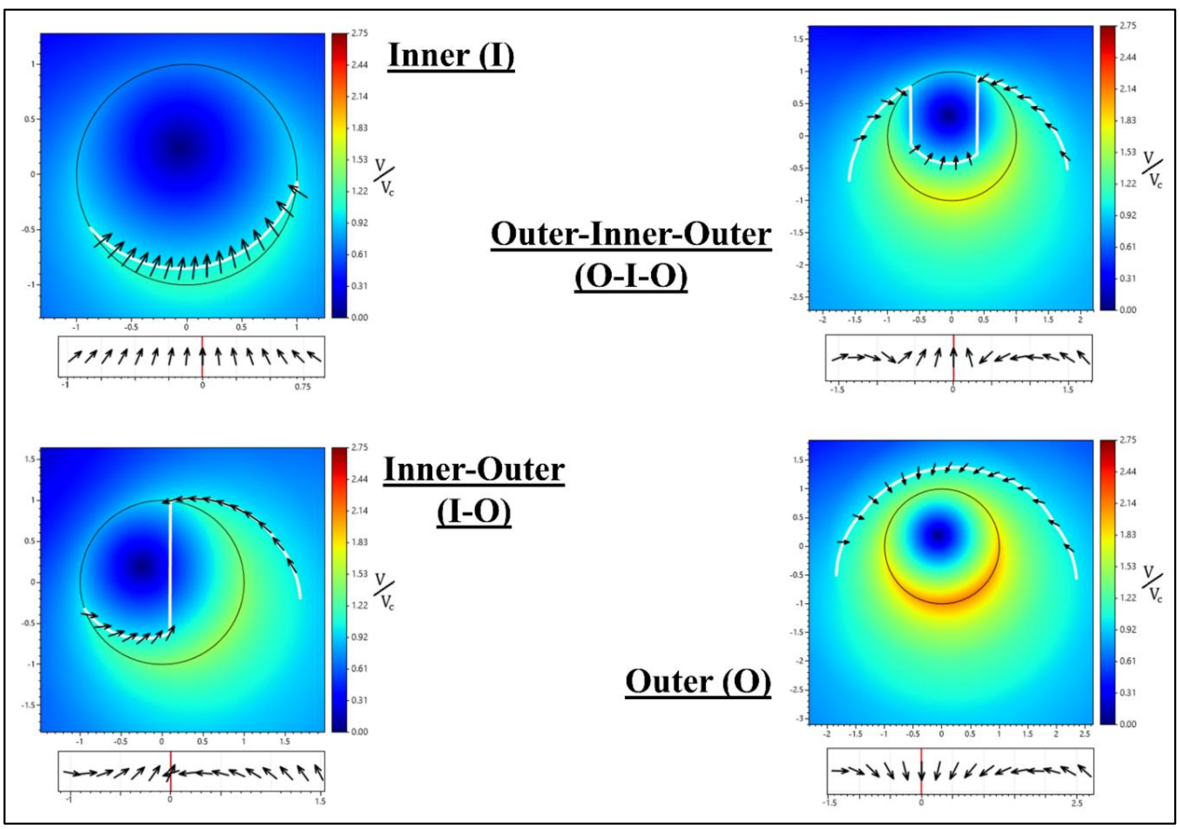
- In general, treefall patterns depend on the specific vortex model profiles, and the ratios between the radial, tangential (rotational), translational, and critical tree failure velocities (V_r, V_t, V_s, V_c).

- This module fixes $V_c = 1$ to limit the ratios of (V_r, V_t, V_s, V_c) and make everything relative to V_c.

- Patterns are displayed with a fixed number of sampled vectors (15) for consistency between different patterns.

- The radius of maximum windspeed (R_{max}) for a fixed number of sampled vectors, only stretches the length of the pattern (vortex), it does not affect the vectors themselves.

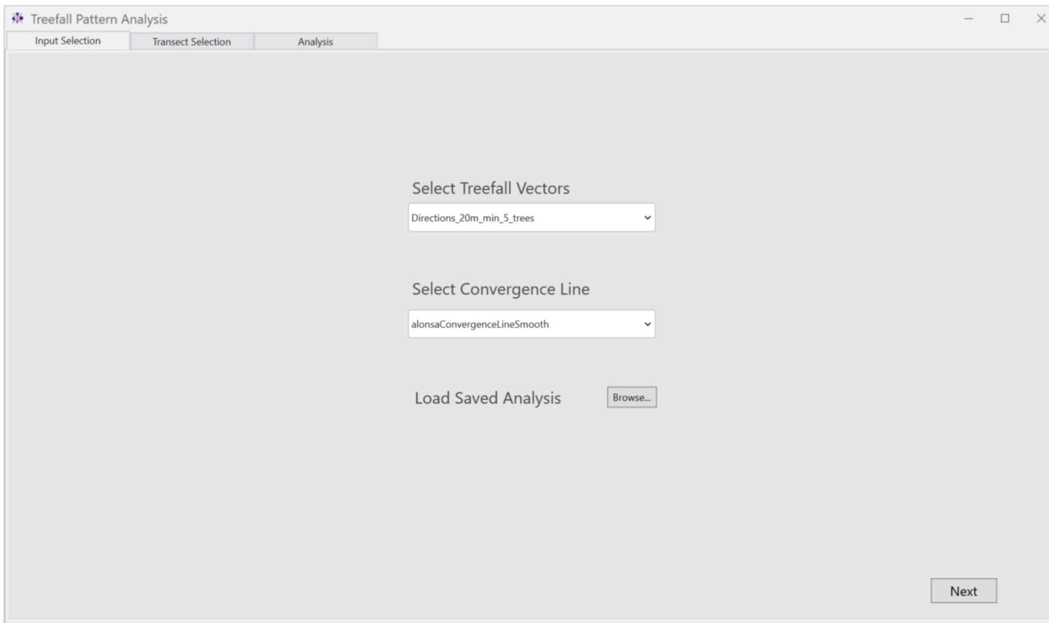
- In general, there are four main types of treefall patterns which can be classified by examining where in the vortex the trees will fail relative to the tornado's radius of maximum windspeed (see paper for more details).



Input Selection

When the “Pattern Analysis” module is opened you will be presented with the “input selection” tab. To start using the software, you must select the treefall vectors and convergence line you wish to use. These must be either feature layers or shapefiles loaded into the ArcGIS Pro project. Once selected, proceed to the next tab by pressing the “Next” button in the bottom right.

An example of area-average treefall directions and a convergence line (black line) can be seen on the title page.



Treefall Vectors

The treefall vectors should be formatted as a polyline feature or shape file, where each vector is a two-point line, treated as a vector from the first point to the second point. It is recommended to use area-averaged treefall vectors if you have more than a few thousand trees for performance of the UI. As discussed in the introduction, CSSL utilizes an automated software to determine these directions and produce the corresponding polyline shapefile.

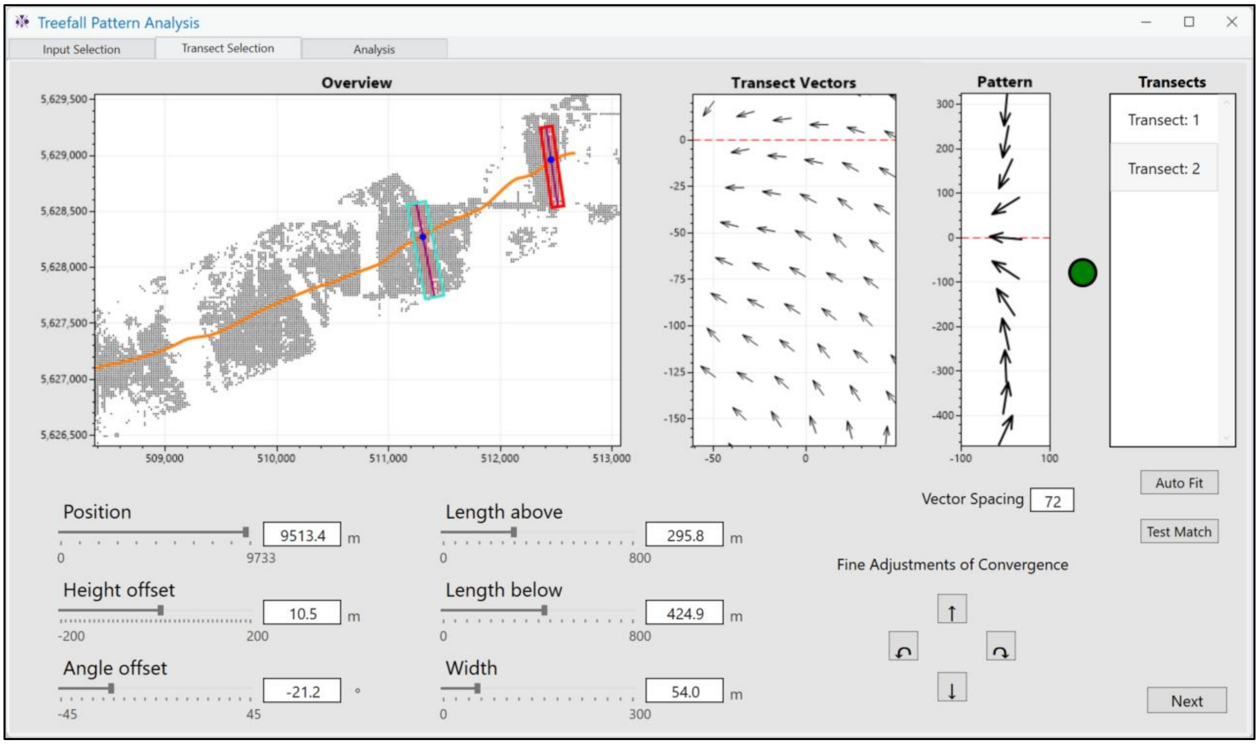
Convergence Line

The convergence line should also be formatted as a polyline feature or shape file, containing a single polyline which describes the center of the convergence of the treefall directions, following in roughly the direction of the tornado’s motion. An estimate for the convergence line should be drawn manually as best as possible given visible treefall direction (it can be adjusted/redrawn)

If you have previously saved an analysis, it can be loaded using the “browse” button. This is discussed more in the “Results” section of the user manual. The save must be loaded on the same ArcGIS project containing the same files used for the vectors/convergence line.

Transect Selection

The "Transect Selection" section is where you fit transects and observe the corresponding treefall pattern created by the vectors inside the transects. The objective here is to find all the district treefall patterns visible in the treefall vectors that look similar to the four types shown in our paper.



1. In order to add a new transect, right click the "Transects" box on the right side and select "New". By adjusting the sliders, you can change the position, size, and orientation of the transect. By adjusting the "Vector Spacing" value you can change the size of the boxes used to calculate the averaged vector used to convert the vectors inside the transect into the pattern.

2. As you adjust the transect, the UI consists of three main graphs. The "Overview" graph shows all the treefall vectors, the convergence line and the placed transects. The "Transect Vectors" graph shows all the vectors currently inside the selected transect. The "Pattern" graph shows the treefall pattern constructed based on the vectors in the selected transect.

3. (Under development) Besides manually placing a transect, clicking on the "Auto Fit" button will attempt to automatically find good places to put transects which are roughly the same size as the currently selected transects (you must first add a transect and size it appropriately).

Good Patterns

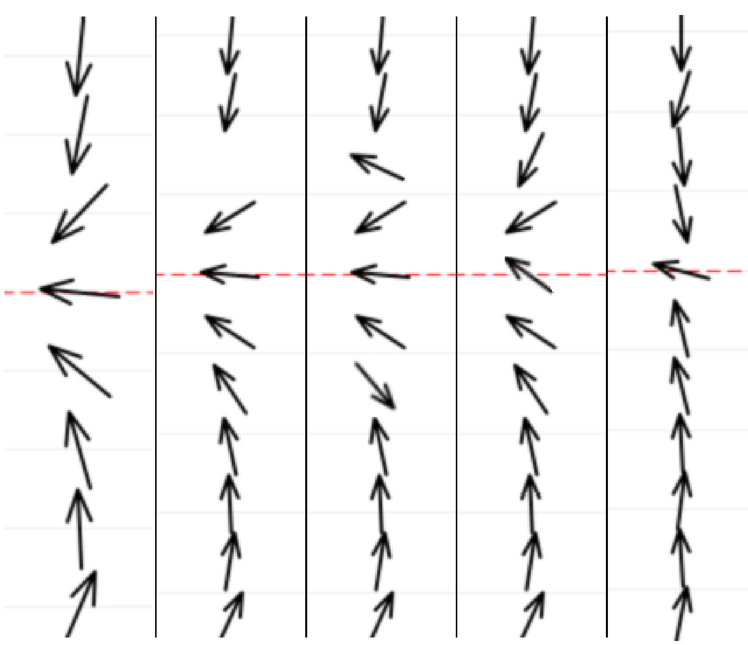
In order to ensure that the transect you've placed will give a good pattern similar to those that can be simulated, it should meet the following criteria:

- Pattern should match one of the four types (Inner, Outer, Inner-Outer, Outer-Inner-Outer)
- 10 – 20 Vectors (less than 10 may not capture the full pattern, more than 20 is unnecessary)
- Pattern extends most of the visible damage width, with no missing vectors
- Vectors should converge towards the center (dotted red line in pattern graph)
- The center vector should be close to parallel with the center (dotted red line) except for in the case of Inner-Outer patterns
- Pattern vectors should rotate towards the centre in the same fashion as the four main types. From top to bottom vectors should rotate clockwise for Outer sections and counterclockwise for Inner sections.

If all these criteria are met, then the circle between the Pattern and Transect selection box will turn from red to green. However, this is not a hard requirement for a good enough pattern. The automated transect fitter attempts to find patterns that meet all the above criteria. The "Test Match" button can also be used to compare your pattern to the closest simulated pattern.

For examples of good patterns use the "Custom Model Analysis" tool described above.

Examples of Bad Patterns



From Left to Right:

Two few vectors

Missing vector

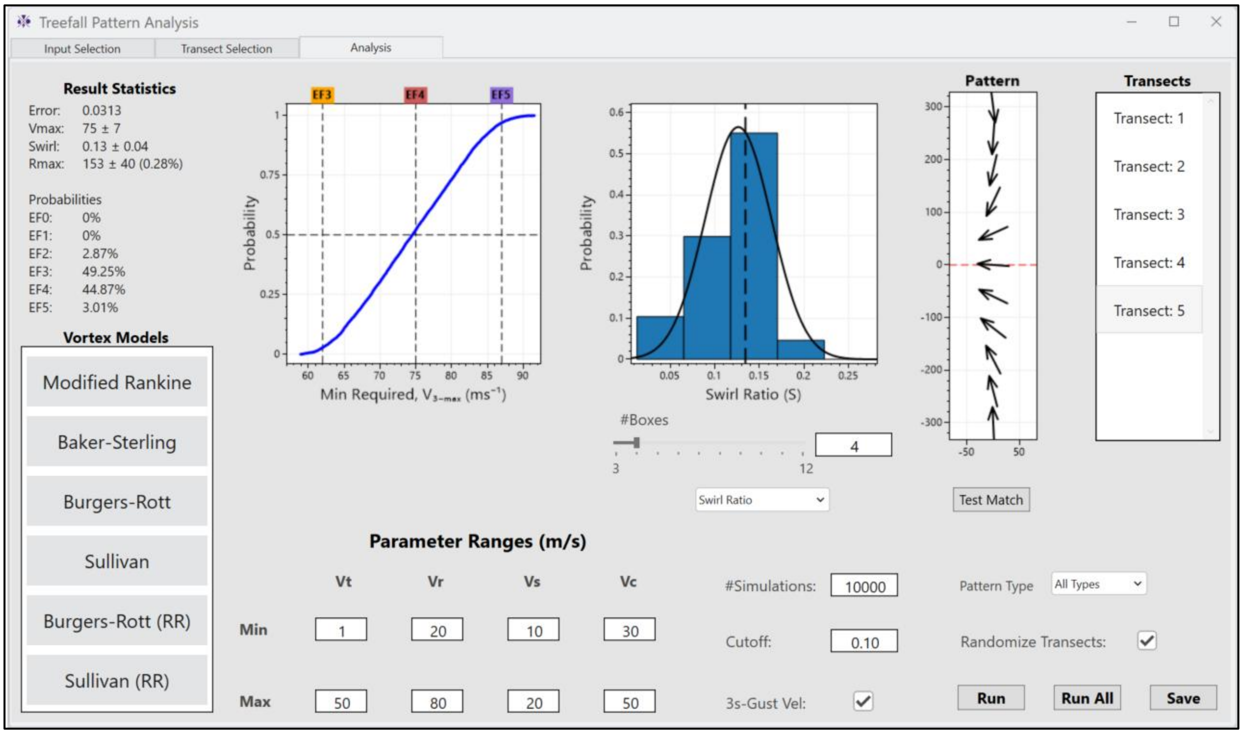
Vectors not all converging

Center vector not parallel to center

Vectors too similar, not rotating

Analysis and Results

Once transects have been fit, proceed to the Analysis tab so they can be analyzed.



1. Start by entering the estimated ranges of V_c and V_s. The ranges of V_r and V_t can also be adjusted, although this is not recommended. Do not enter values less than or equal to 0. Max values must be greater than the min values. The max V_s must be greater than the min V_c value.

2. The various vortex models used in the simulation can be toggled off/on by selecting them in the “Vortex Models” box on the left.

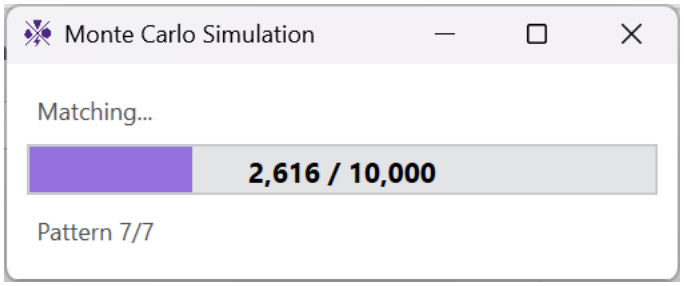
3. The number of simulations to perform in the Monte Carlo simulation can be changed using the “#Simulations” box. It is recommended to use 10000 for results and at least 1000 for testing.

4. The “Cutoff” threshold is used to determine the maximum error between the observed and simulated treefall patterns. Any simulated vortex which produces a treefall pattern with an error below this threshold is considered a plausible match for the true vortex. (0.1 recommended)

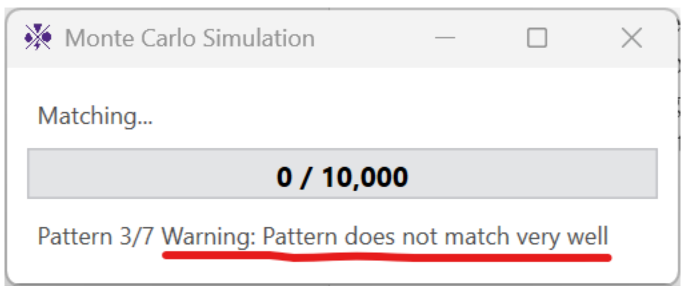
5. The “3s-Gust Vel” toggles using the max 3s-Gust velocity or the max instantaneous velocity. The “Randomize Transects” toggle applies random small adjustments during each iteration.

6. The “Pattern Type” drop-down can be used to restrict the types of simulated patterns. Importantly, selecting the “Not Outer” option changes the windspeed from the minimum to the median result in each individual iteration of the Monte Carlo Simulation as an upper bound windspeed exists for Non-Outer types but not for Outer type when R_{max} is unknown.

Once all the parameters and settings have been entered, clicking the "Run" button will run the Monte Carlo simulation on the selected transect. The "Run All" button can be used to run all the transects in order. The simulation can be cancelled by closing the progress window.



Important Note: if the best-matching simulated pattern for the selected transect does not have a matching error below the cutoff threshold, then the simulation will get stuck (as no plausible patterns will be found). Generally, if the best-matching error is not less than half the cutoff threshold then the selected transect is probably not placed effectively and should be adjusted. The most common reason for the simulation running slowly is poorly fit transects. The progress window will display a warning if the pattern does not match well:



Once complete, the results on the simulation are shown the top left panel and two central graphs. If you wish to save your transects/results, use the "Save" button in the bottom right corner. This will save the transects, settings, and results in a JSON format which can be loaded into the software from the "Input Selection" tab.

Troubleshooting

1. After installation, the add-in is not present or doesn't open properly.
    a. Make sure you've updated ArcGIS Pro to the latest version
    b. Try closing ArcGIS Pro and reinstalling the add-in

2. The custom model graphs are slow or lagging
    a. Try reducing the "Number of Samples" to reduce the calculations required.

3. I can't add/remove nodes from the custom model profile graphs
    a. You must be dragging a node while pressing the "n" or "d" keys
    b. Try adjusting the parameter sliders to reset the graphs

4. Errors loading the selected treefall vectors or convergence line
    a. Make sure you've followed the correct file format in the "Input Selection" section

5. Attemping to load a saved analysis doesn't work or causes the software to crash
    a. Make sure you're opening the save in the same ArcGIS Pro project and that the same vectors/convergence line layers are present in the project
    b. The save file may be corrupted/from a previous version of the software

6. The Overview graph is very slow or lagging
    a. The graph may lag if you have too many treefall vectors, try using larger area-averaged treefall directions

7. I can't manually find any transects which give good patterns and the auto transect fitter doesn't work or doesn't find any transect locations
    a. Try adjusting the convergence line to be more accurate
    b. Try adjusting the size of the reference transect used for the auto transect fitter
    c. Not all tornado events are suitable for treefall pattern analysis. Possible explanations include a multi-vortex tornado, mixture of downburst and tornado damage, a fast-travelling tornado with weak trees ($V_S > V_c$), terrain effects.

8. The simulation is stuck or takes forever to run
    a. Make sure you have a good pattern. See the "Important Note" in the Analysis section
    b. Make sure you've entered the right parameters or try increasing parameter ranges
    c. If your PC is on the slower side, you can try the following to simplify the analysis
        i. Select the Pattern Type instead of using "All Types"
        ii. Reduce the number of simulations needed (#Simulations)
        iii. Select only the Baker-Sterling model (fastest model)
        iv. Disable "Randomize Transects"